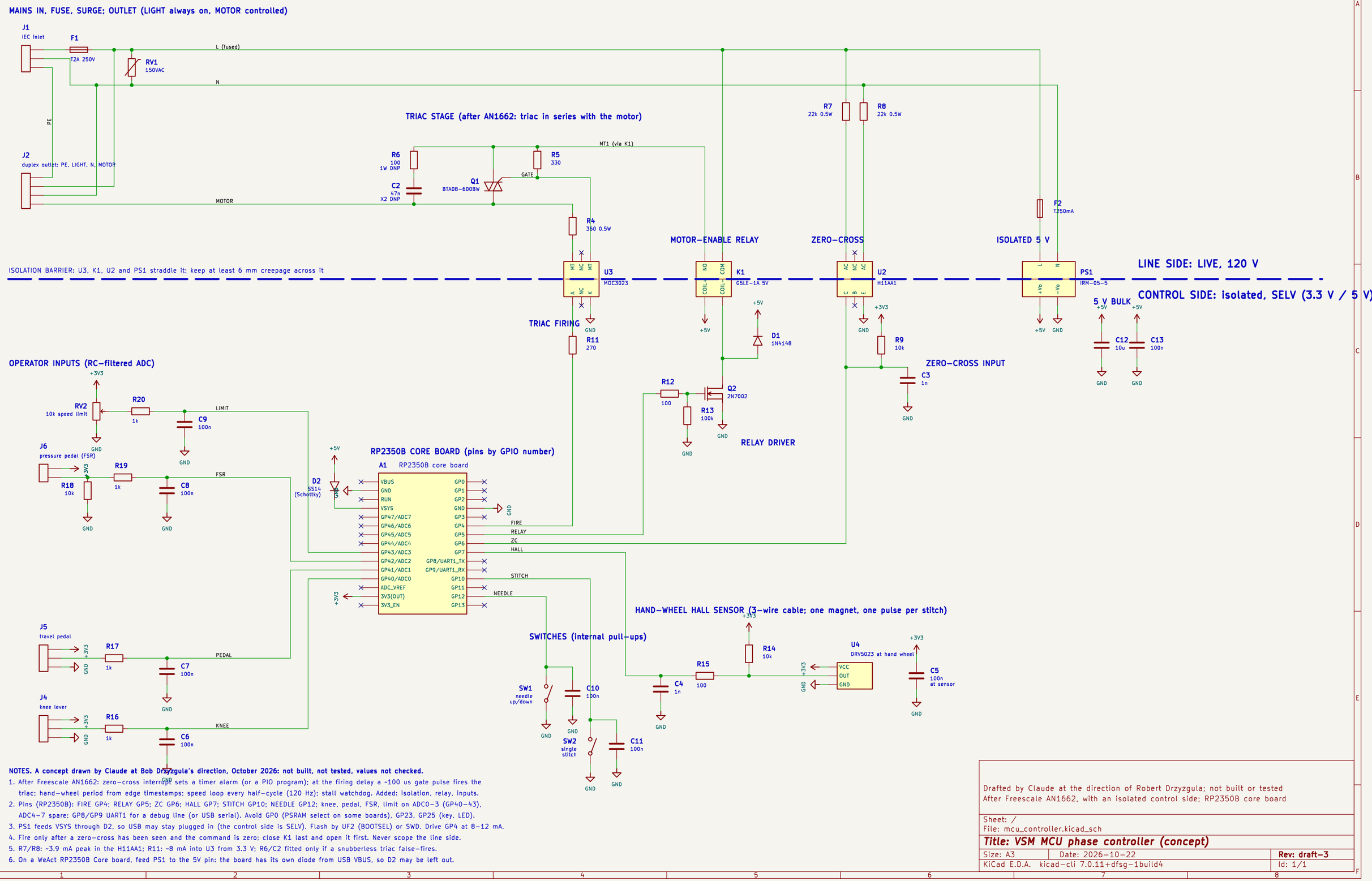

MAINS IN, FUSE, SURGE; OUTLET (LIGHT always on, MOTOR controlled)
J1
IEC inlet
F1
T2A 250V
RV1
150VAC
L (fused)
N
PE
J2
duplex outlet: PE, LIGHT, N, MOTOR
MOTOR
TRIAC STAGE (after AN1662: triac in series with the motor)
MT1 (via K1)
R6
100
1W DNP
C2
47n
X2 DNP
Q1
BTA08-600BW
R5
330
GATE
R4
360 0.5W
R7
22k 0.5W
R8
22k 0.5W
F2
T250mA
MOTOR-ENABLE RELAY
ZERO-CROSS
ISOLATED 5 V
ISOLATION BARRIER: U3, K1, U2 and PS1 straddle it; keep at least 6 mm creepage across it
LINE SIDE: LIVE, 120 V
CONTROL SIDE: isolated, SELV (3.3 V / 5 V)
U3
MOC3023
K1
G5LE-1A 5V
U2
H11AA1
PS1
IRM-05-5
5 V BULK
C12
10u
C13
100n
TRIAC FIRING
R11
270
D1
1N4148
R9
10k
ZERO-CROSS INPUT
C3
1n
OPERATOR INPUTS (RC-filtered ADC)
RV2
10k speed limit
R20
1k
LIMIT
C9
100n
R12
100
Q2
2N7002
R13
100k
RELAY DRIVER
J6
pressure pedal (FSR)
R19
1k
R18
10k
C8
100n
FSR
RP2350B CORE BOARD (pins by GPIO number)
A1 RP2350B core board
D2
SS14
(Schottky)
FIRE
RELAY
ZC
HALL
STITCH
NEEDLE
J5
travel pedal
R17
1k
C7
100n
PEDAL
J4
knee lever
R16
1k
C6
100n
KNEE
SWITCHES (internal pull-ups)
SW1
needle up/down
C10
100n
SW2
single stitch
C11
100n
HAND-WHEEL HALL SENSOR (3-wire cable; one magnet, one pulse per stitch)
R15
100
R14
10k
C4
1n
U4
DRV5023 at hand wheel
C5
100n
at sensor
NOTES. A concept drawn by Claude at Bob Drzyzgula's direction, October 2026: not built, not tested, values not checked.
1. After Freescale AN1662: zero-cross interrupt sets a timer alarm (or a PIO program); at the firing delay a ~100 us gate pulse fires the triac; hand-wheel period from edge timestamps; speed loop every half-cycle (120 Hz); stall watchdog. Added: isolation, relay, inputs.
2. Pins (RP2350B): FIRE GP4; RELAY GP5; ZC GP6; HALL GP7; STITCH GP10; NEEDLE GP12; knee, pedal, FSR, limit on ADC0-3 (GP40-43), ADC4-7 spare; GP8/GP9 UART1 for a debug line (or USB serial). Avoid GP0 (PSRAM select on some boards), GP23, GP25 (key, LED).
3. PS1 feeds VSYS through D2, so USB may stay plugged in (the control side is SELV). Flash by UF2 (BOOTSEL) or SWD. Drive GP4 at 8-12 mA.
4. Fire only after a zero-cross has been seen and the command is zero; close K1 last and open it first. Never scope the line side.
5. R7/R8: ~3.9 mA peak in the H11AA1; R11: ~8 mA into U3 from 3.3 V; R6/C2 fitted only if a snubberless triac false-fires.
6. On a WeAct RP2350B Core board, feed PS1 to the 5V pin: the board has its own diode from USB VBUS, so D2 may be left out.
Drafted by Claude at the direction of Robert Drzyzgula; not built or tested
After Freescale AN1662, with an isolated control side; RP2350B core board
Sheet: /
File: mcu_controller.kicad_sch
Title: VSM MCU phase controller (concept)
Size: A3
Date: 2026-10-22
Rev: draft-3
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 1/1